Project Plan: Smart Workspace Allocation (Knowit)

1. Problem definition

Many organizations, including Knowit and its clients, are facing a major challenge regarding the allocation of their workspaces. Following downsizing or relocations, offices experience days with a shortage of space and other days with unused surplus. Traditional booking tools lack flexibility and fail to reflect actual usage patterns or employee needs. The company's need is therefore to better understand and optimize workspace usage in a flexible, data-informed manner. However, this project has a strict and fundamental constraint: the solution must absolutely respect employee privacy and avoid creating any sense of surveillance (a privacy-friendly approach).

2. Measurable goals

Our main goal is to solve the daily office occupancy imbalance in order to reduce time wasting and employee stress, while providing reliable statistics to management. Specifically, as shown in our **Effort / Impact matrix** below, our deliverables and success criteria are as follows:

Delivery of a data-driven concept: Design of a solution integrating an occupancy **"Heat-map"** and an analysis module (**"Predicted / real capacity stats"**).

Reduction of search time: The tool must drastically reduce the time an employee spends looking for a suitable desk in the morning.

Privacy compliance: 100% anonymization in data collection to respect Knowit's guidelines.

User validation: Achieving a high adoption rate and positive feedback during our **"User tests"**, **"Interviews"**, and **"Research & survey"** phases.

3. Graphical description of the problem understanding

To clearly outline the stakes of this project, we modeled our understanding of the problem using three analysis tools (First Diamond methods).

First, as seen on the **Stakeholder Map** below, the problem directly affects a core ecosystem consisting of the employees (the core), colleagues, HR, and the employer.

[Insert Stakeholder Map image here - image_a6f840.png]

Second, the **Cause-Effect Tree** perfectly illustrates the consequences of this workspace issue on the employee: uncertainty generates interruptions and time wasting, which inevitably leads to work inefficiency ("Cannot complete task").
[Insert Cause-Effect Tree image here - image_a6f845.png]

Finally, to strategically define our action plan, we categorized our potential solutions. As shown on the **Effort / Impact Matrix** below, we decided to exclude individual named-desk booking systems (too rigid) to focus on Quick wins (structure work) and Major tasks (Heat-map, predictive stats, research & survey).
[Insert Effort/Impact Matrix image here - image_a6f862.png]

4. Approaches

To meet the need without compromising privacy, our approach will focus on "macro" management (by zone or by team) rather than individual tracking. The conceptual architecture of our solution will be structured around aggregated data collection and the rendering of an occupancy "Heat-map". *(The exact technical definition of the components and the prototype is intentionally put on hold at this stage of the project, while we refine our research).*

5. Project methods and development tools

Project Method: We will adopt an Agile project management methodology, which is recommended to adapt to continuous feedback. The work will be iterative, with regular synchronization meetings.

Development Tools: For now, no specific technical tools are being used in this research and design phase. In the future, during the implementation phase, we plan to use:

- **VS Code** for programming and technical development.
- **Smartphones** to conduct user tests in real conditions (mobile interface).
- **Google Forms** to create and distribute our questionnaires during the *Research & survey* phase.